



The University of Texas at Austin
McCombs School of Business

Banking on AI for Customer Retention

Predicting customer exit with artificial neural networks

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Introduction to Neural Networks · April 2024

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PROJECT

04

Introduction to Neural Networks

Fifteen feed-forward network configurations in TensorFlow/Keras — optimizers, dropout, class weighting, SMOTE, regularization and thresholding — to identify bank customers who will leave.

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Executive summary

20.4%

of 10,000 customers have exited the bank

15

network configurations trained and compared on validation recall

0.93

test recall on exiting customers — selected model (binned features, class-weighted, threshold 0.3)

0.90

test recall for the runner-up, which keeps half of non-churners correctly classified

Key findings

Germany, women, inactive members and the 45–55 age band exit at well above the base rate. Germany's exit rate is double France's and Spain's.

Every high-recall model pays in precision. At threshold 0.3 the selected network flags most customers as at-risk; the business must decide what a false alarm costs before deploying.

The data lacks the strongest predictors. Transaction patterns, credit utilization, complaints and satisfaction scores would give a more complete view than any architecture change.

Actionable insights and recommendations

01

Target retention in Germany

A 32% exit rate on 2,509 customers. Loyalty programs, personalized offers and service improvements specific to the region.

02

Learn from Spain

Investigate why Spain retains best (17%) and replicate what transfers to Germany and France.

03

Tailor offers to female customers

Fewer women in the book, but a 25% exit rate against 16% for men.

04

Re-engage inactive members

Inactive members exit at 27% against 14% for active ones. Engagement programs and reactivation campaigns aimed at falling usage.

05

Focus on the middle-age segment

Exit peaks at 49% for ages 45–55. Products and service suited to that stage of life.

06

Advise high-balance customers

Exiting customers carry the higher median balance (109k vs 92k). Personalized financial planning and balance-tier loyalty incentives.

07

Collect what the model lacks

Credit utilization, transaction frequency and variability, service interactions, NPS, complaint history.

08

Set the threshold with the business

Threshold 0.3 buys recall at the cost of precision; the right point depends on the cost of an unnecessary retention offer.

Business problem and solution approach

Problem

- About 20% of customers have stopped using the bank's services. Build a model that correctly identifies the customers who will exit, so retention effort can be directed at them.

Primary metric: recall on the exit class. Missing a customer who leaves costs more than contacting one who would have stayed.

Approach

- EDA — univariate and bivariate against the exit flag
- Preprocessing: feature removal, quantile binning, dummy variables, normalization, SMOTE
- Sequential Keras networks: optimizer, dropout, initialization, batch normalization, class weights, regularization, activation, decision threshold
- Fifteen configurations compared on validation recall; four carried to test
- Final model selection and business recommendations

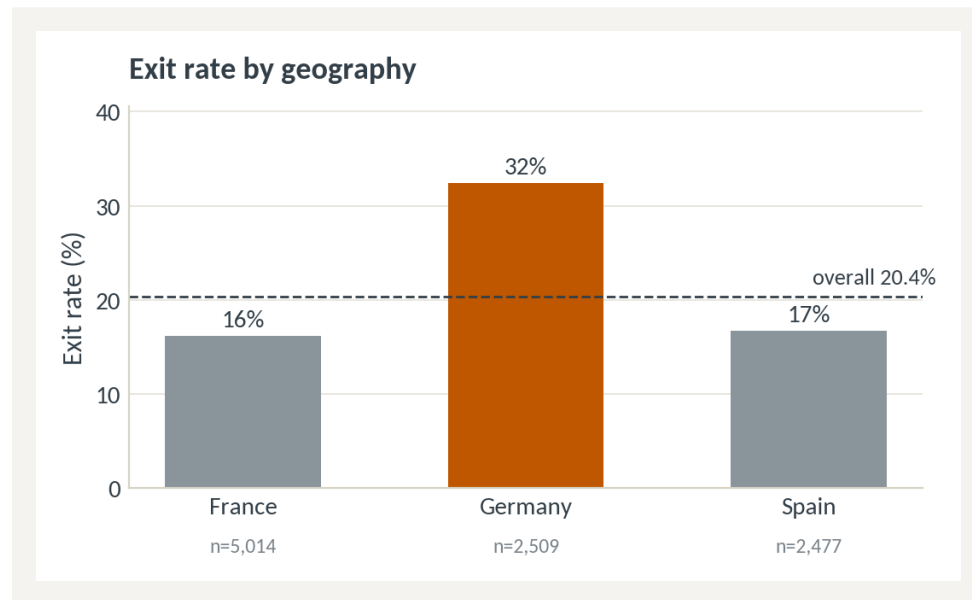
EDA results

Who leaves

Geography and gender

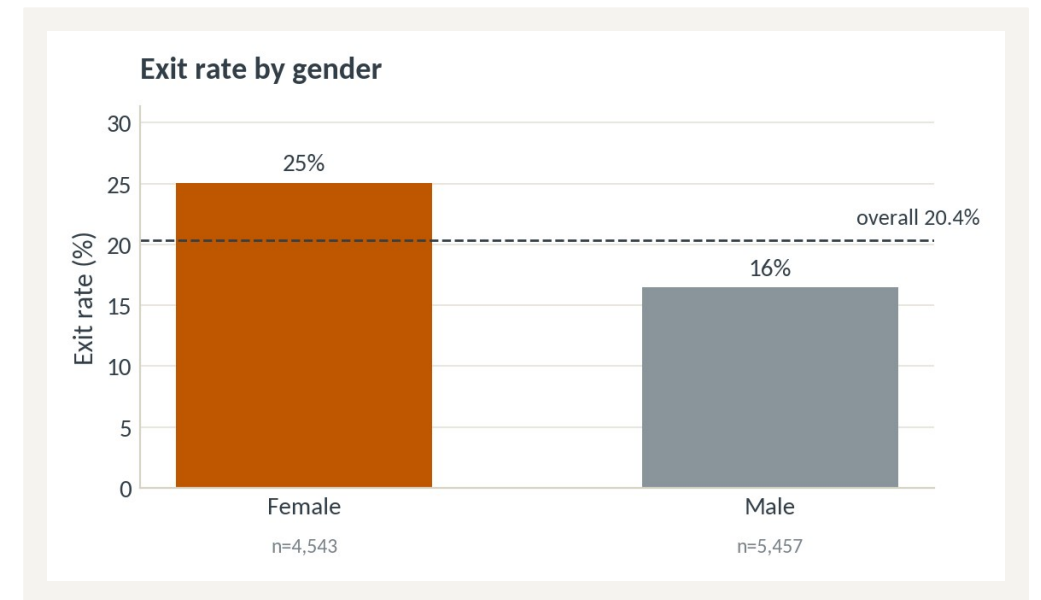
Germany exits at twice the rate of France or Spain

- Despite the smaller customer base, Germany has the higher exit rate in both absolute and proportional terms.



Women exit more often than men

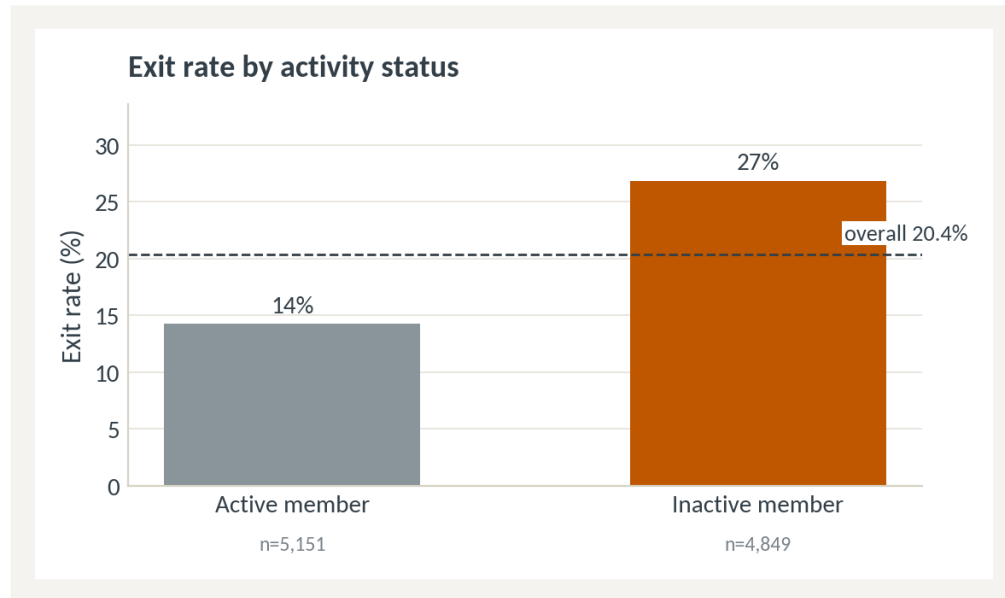
- Fewer female customers, yet a higher exit rate: 25% against 16%.



Activity and age

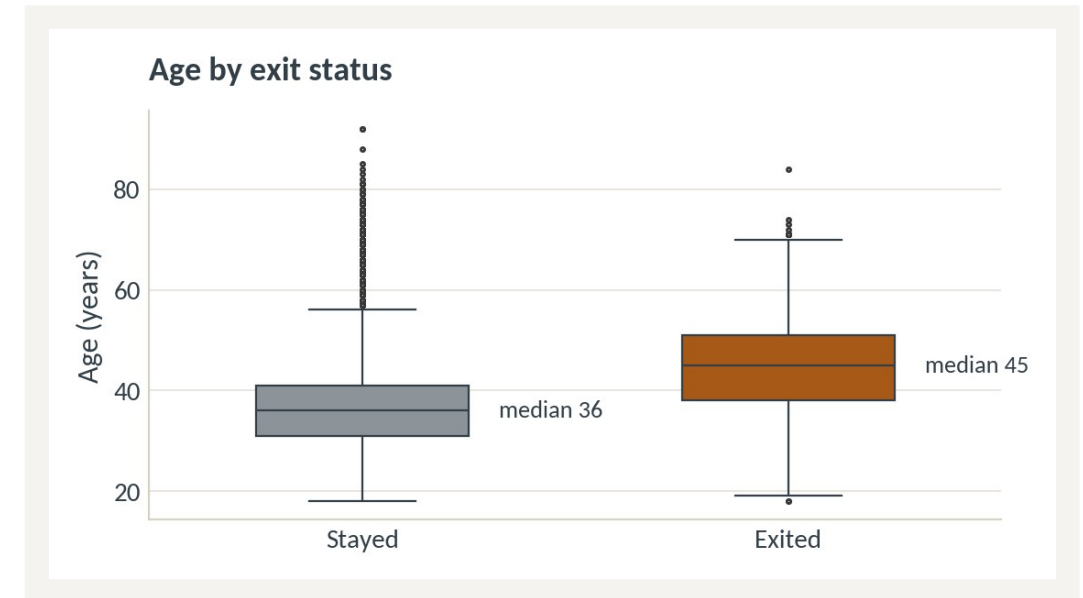
Inactive members leave at nearly twice the rate

- 27% of inactive members have exited against 14% of active ones.



Exiting customers are older

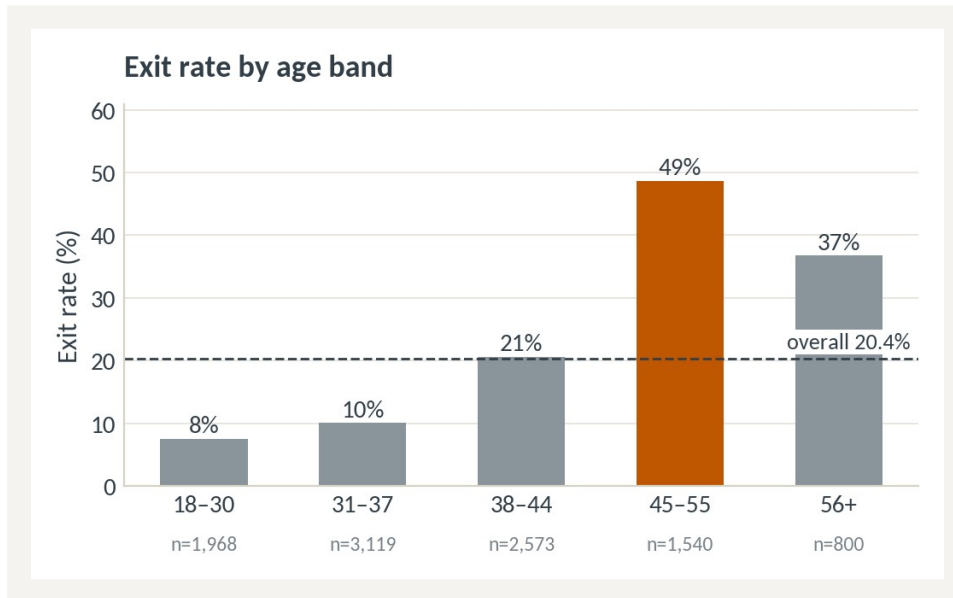
- Median age 45 against 36. The stayers carry the older outliers — customers well above the average age who remain.



Age band and balance

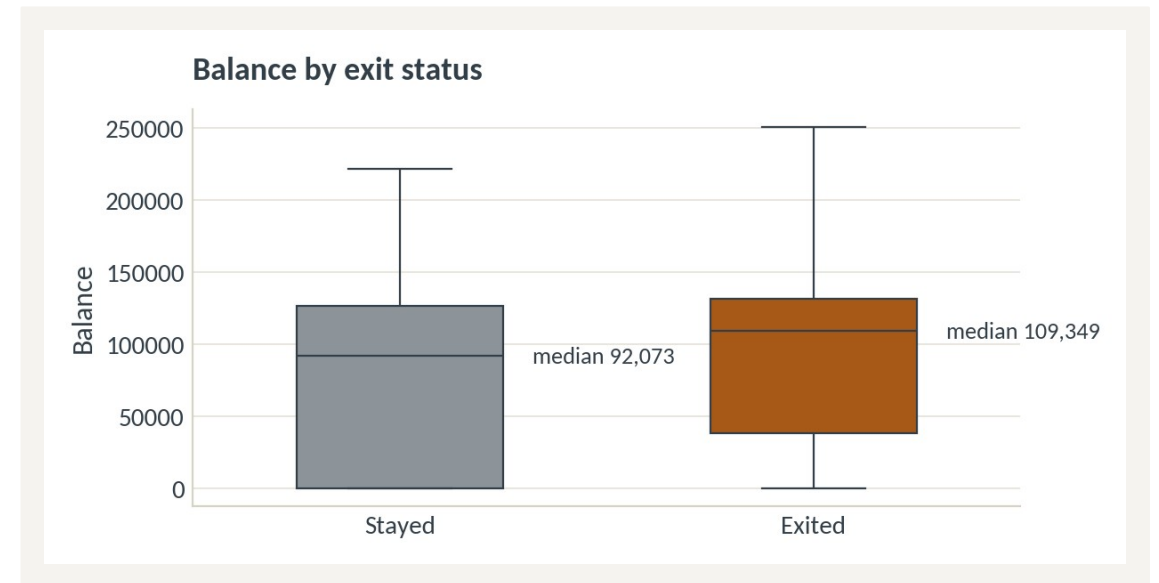
Exit peaks in middle age

- The 45–55 band exits at 49%; 56 and over at 37%; under 38 at 10% or less.

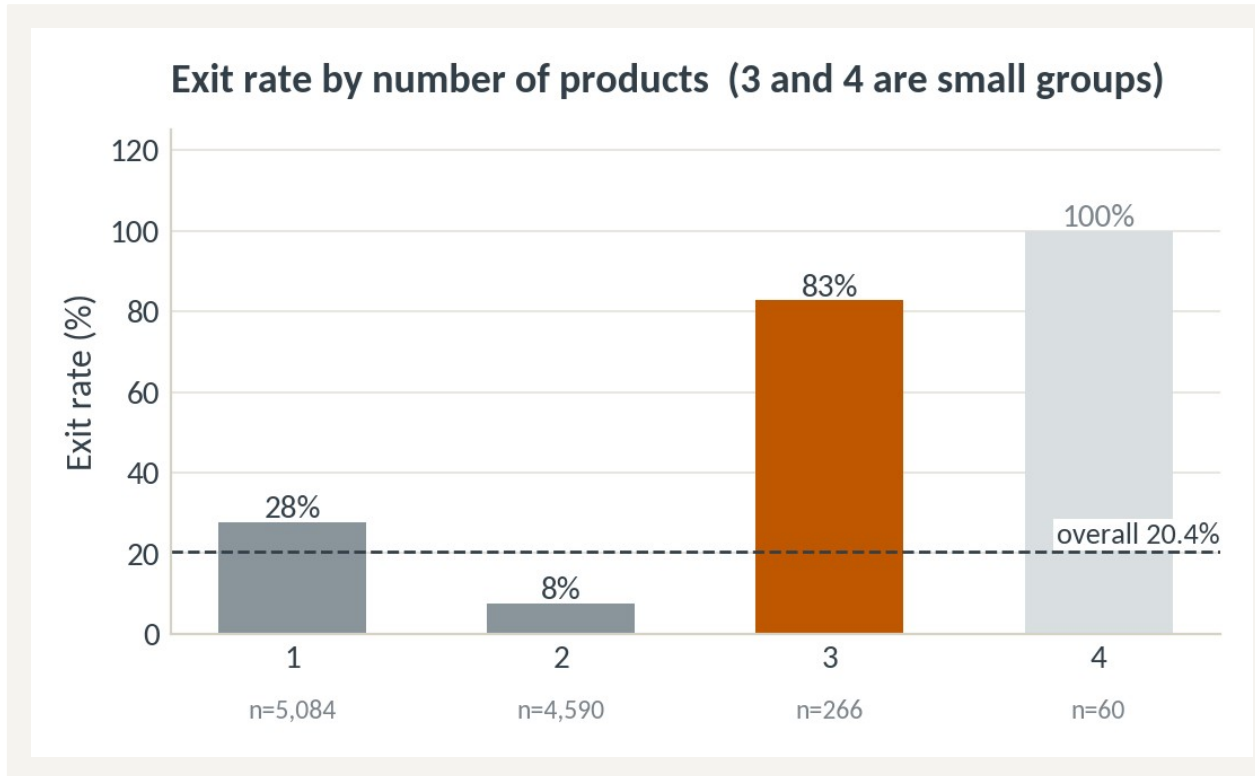


Exiting customers hold slightly more

- Median balance around 109k for those who left against 92k for those who stayed; both groups center near the 100k mark.



Number of products



Two products is the stable position: 8% exit.

One product: 28%. The bulk of the book (5,084 customers) sits here.

Three and four products exit at 83% and 100%, but these are small groups (266 and 60) — a signal worth checking, not a finding to act on alone.

Data preprocessing

Cleaning and features

Duplicates and missing values. None.

Outliers. Present in age among older customers who stayed; retained as a meaningful group.

Removed identifier columns with only unique values (row number, customer ID, surname).

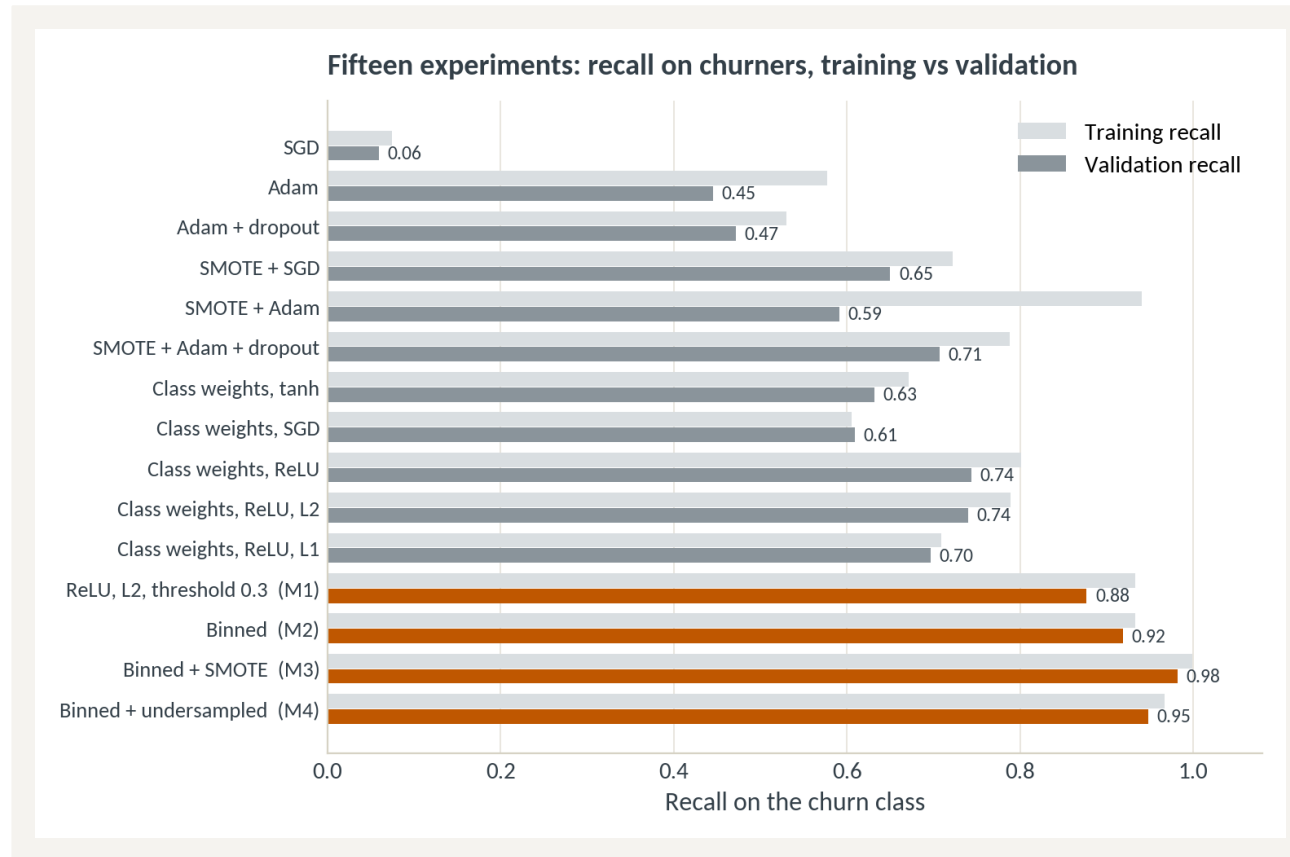
Quantile binning of age and balance to expose which bands exit.

Preparation for modeling

- Train, validation and test split; test set of 2,000 customers.
- Dummy variables for geography and gender.
- Standard-scaled numeric features.
- SMOTE oversampling and class weights as alternative treatments for the 80/20 imbalance.

Model experiments and comparison

Fifteen configurations, one metric



Plain SGD fails (recall 0.06): the network predicts the majority class.

Adam, dropout and class weights lift validation recall to the 0.6–0.74 range.

SMOTE + Adam overfits — training recall 0.94, validation 0.59.

The decisive change is the threshold. Lowering the decision threshold from 0.5 to 0.3 on the class-weighted ReLU/L2 network takes validation recall from 0.74 to 0.88.

Binning adds a further three points (0.92); SMOTE and undersampling on binned features push recall to 0.95–0.98 — at a precision cost visible on test.

Test-set performance of the four finalists

Model (all: Adam, dropout, class weights, He init., batch norm., lr 1e-4, ReLU, L2, threshold 0.3)	Recall · exit	Recall · stay	Precision · exit	F1 · exit	Accuracy
M1 Unbinned features	0.90	0.51	0.32	0.47	0.59
M2 Binned features (selected)	0.93	0.34	0.26	0.41	0.46
M3 Binned + SMOTE	0.99	0.15	0.23	0.37	0.32
M4 Binned + undersampled	0.96	0.19	0.23	0.37	0.35

Every finalist trades the stay class for the exit class. M3 catches 99% of exits by flagging 85% of stayers too; M1 keeps half the stayers correctly classified at 90% exit recall.

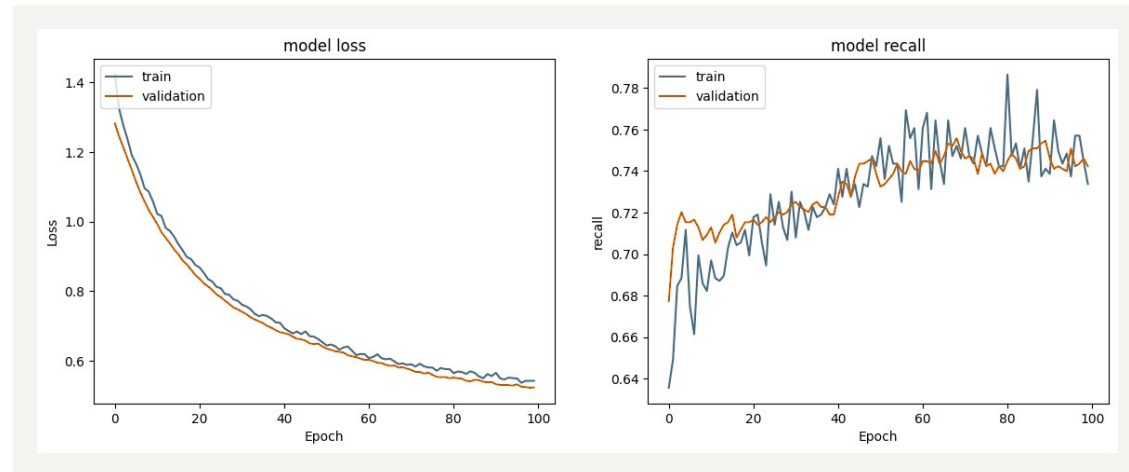
Selection. If the cost of misidentifying a non-churner is very low, M2 is the choice (recall 0.93). If that cost is material, the business should prefer M1, which gives up three points of recall for a far cleaner stay class (0.51 vs 0.34).

Test set: 2,000 customers, 407 of whom exited.

Training dynamics — M1 and M2

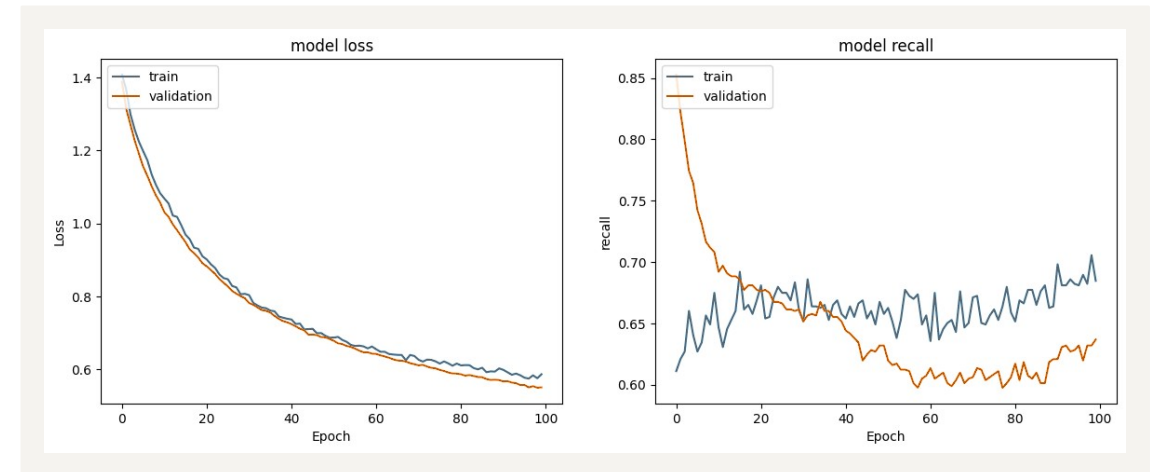
M1 • unbinned

- Loss falls smoothly on train and validation; recall climbs to about 0.74 on both and keeps pace. The model is learning without divergence.



M2 • binned (selected)

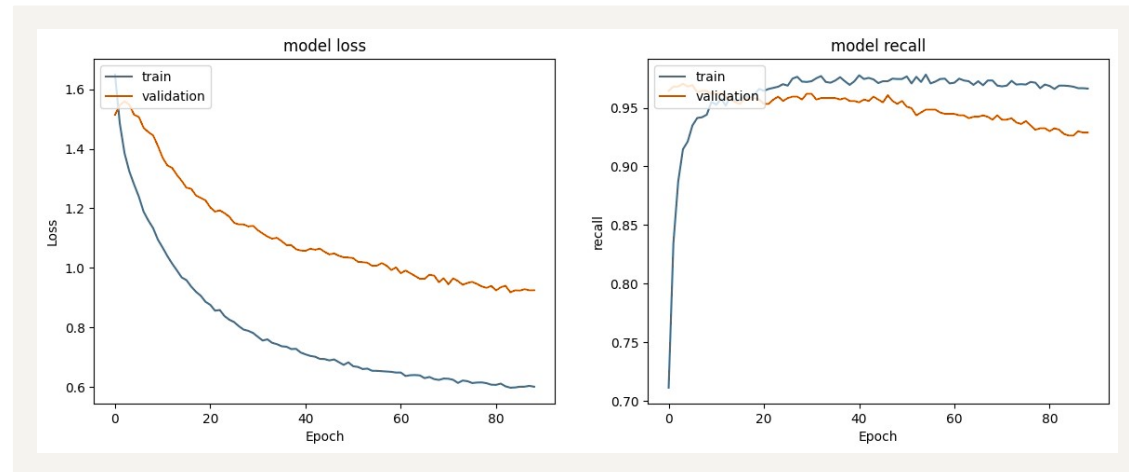
- Loss behaves the same; validation recall drops early and recovers only partly — the higher test recall comes from the threshold, not from a better fit.



Training dynamics — M3 and M4

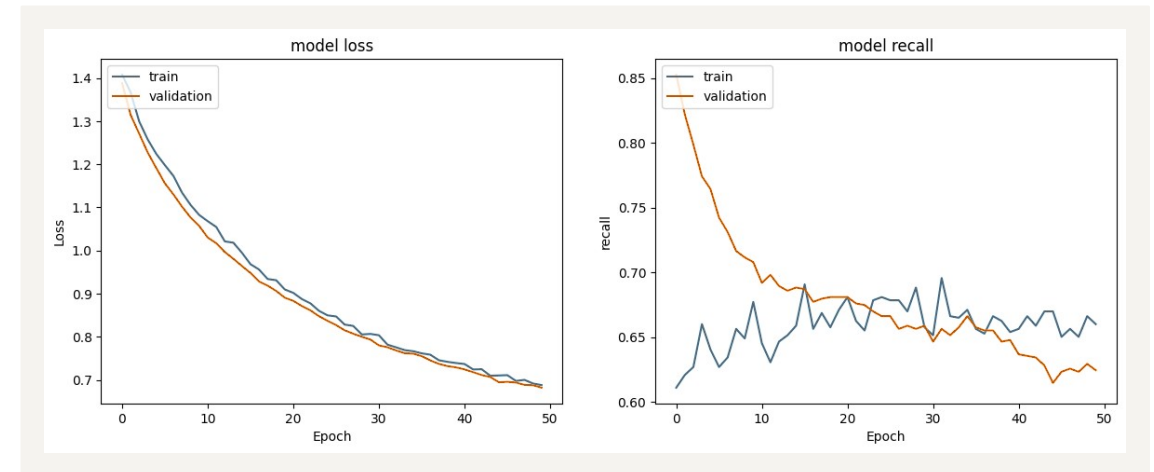
M3 • binned + SMOTE

- Stable curves, but the gap between training and validation loss widens throughout: the synthetic rows are being memorized.



M4 • binned + undersampled

- Loss barely moves (0.57 → 0.53) and validation recall oscillates around 0.64 — thin training data after undersampling.



Final model

Selected model — architecture and training

Architecture (12,161 parameters)

Input layer: 128 neurons, ReLU, He-normal initialization.

Dropout 0.3.

Hidden layer: 64 neurons, ReLU, He-normal; batch normalization.

Hidden layer: 32 neurons, ReLU, L2 regularization (0.01); dropout 0.3.

Output: single sigmoid neuron; predictions thresholded at 0.3.

Training

Optimizer: Adam, learning rate 1e-4.

Loss: binary cross-entropy; recall tracked as the metric.

Class weights to counter the 80/20 imbalance without synthetic data.

Up to 100 epochs, batch size 32, early stopping on validation loss.

Features: binned age and balance alongside the standard-scaled numerics; random state fixed for reproducibility.

What the result means for the bank

Recall 0.93 on test: of 407 customers who left, the model would have flagged 379 in advance.

Precision 0.26: roughly three of every four customers flagged would not have left. A retention offer costs money; a lost customer costs more. Where that line sits is a business decision, and the threshold is the dial.

Threshold 0.5 → 0.3 moved recall from 0.74 to 0.88 on validation; a threshold sweep against a cost model is the natural next step.

Better features beat better architecture. Nothing in the fifteen experiments changed recall as much as the threshold did; the data carries no transaction behavior, which is where churn signal lives.

Appendix

All experiments and data background

All fifteen experiments — recall on the exit class

#	Configuration	Train	Valid.
1	SGD	0.07	0.06
2	Adam	0.58	0.45
3	Adam + dropout	0.53	0.47
4	SMOTE + SGD	0.72	0.65
5	SMOTE + Adam	0.94	0.59
6	SMOTE + Adam + dropout	0.79	0.71
7	Class weights, He init., batch norm., lr 1e-4, tanh	0.67	0.63
8	Same, SGD	0.60	0.61
9	Same, Adam, ReLU	0.80	0.74
10	Same, ReLU, L2	0.79	0.74
11	Same, ReLU, L1	0.71	0.70
12	Same, ReLU, L2, threshold 0.3 (M1)	0.93	0.88
13	Binned features (M2)	0.93	0.92
14	Binned + SMOTE (M3)	1.00	0.98
15	Binned + undersampled (M4)	0.97	0.95

Experiments 1-3: optimizer and dropout on the raw imbalance. Adam learns; SGD does not.

4-6: SMOTE-balanced data. Recall rises; SMOTE + Adam overfits badly.

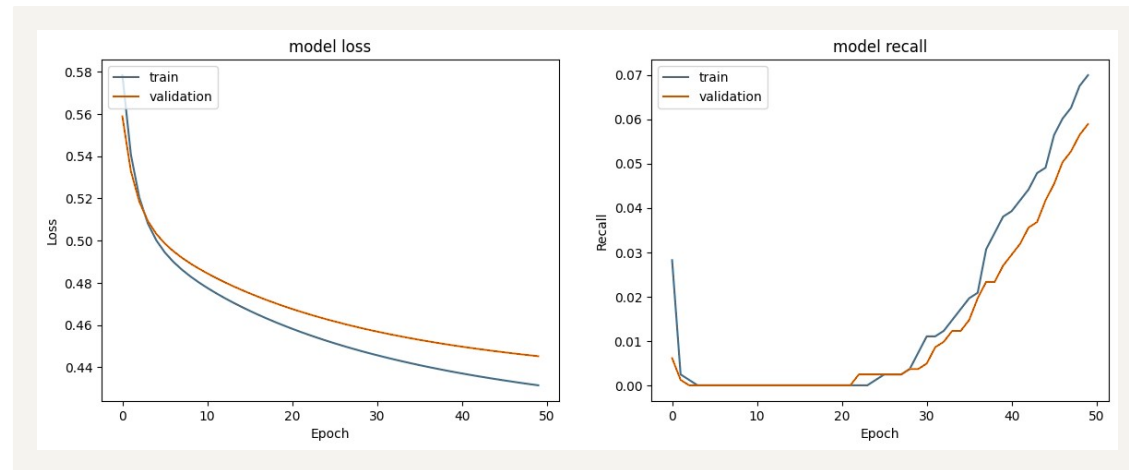
7-11: class weights with He initialization and batch normalization; activation and regularization varied. ReLU beats tanh; L2 ≈ none; L1 hurts.

12-15: decision threshold lowered to 0.3, then binned features and resampling. Recall climbs; precision falls (see test table).

Appendix — optimizer experiments (1–3)

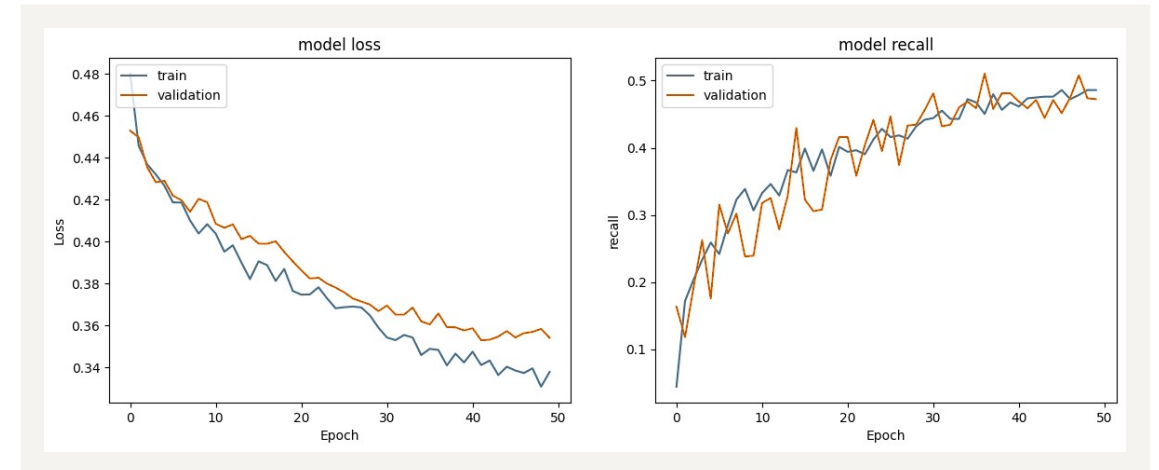
SGD

- Loss falls, but recall on exiting customers never rises above 0.07: the network learns to predict 'stay'.



Adam, then Adam + dropout

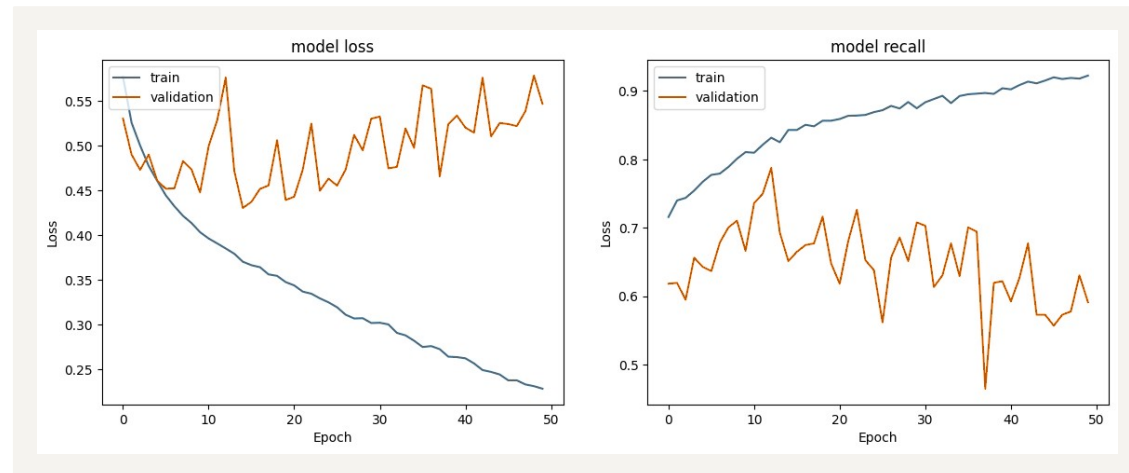
- Adam learns; validation recall diverges from training after epoch 20. Dropout closes the gap and steadies the recall curve.



Appendix — SMOTE experiments (4–6)

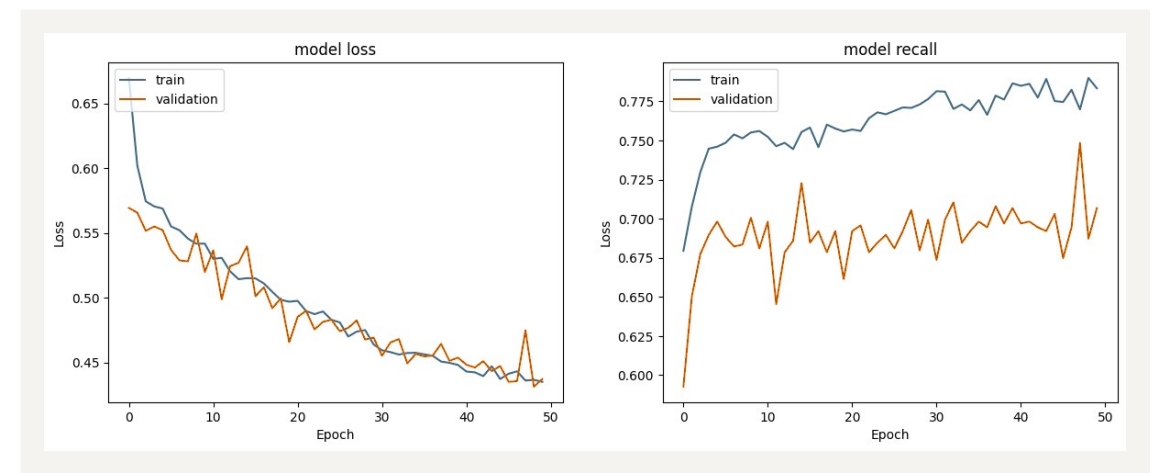
SMOTE + Adam

- Training recall climbs to 0.94 while validation loss rises from epoch 5: the synthetic minority is memorized.



SMOTE + Adam + dropout

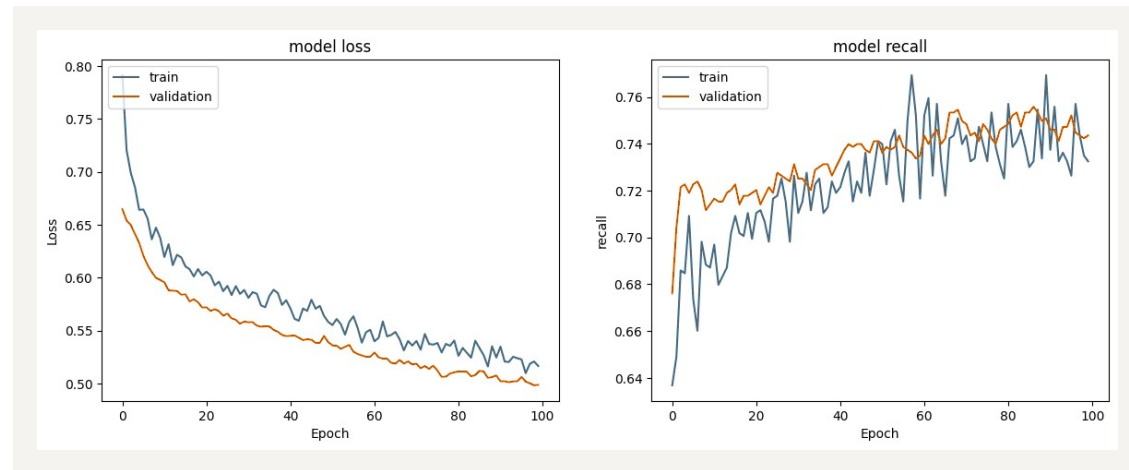
- Dropout restores generalization; validation recall settles at 0.71 with the training curve close above it.



Appendix — class-weight experiments (7–11)

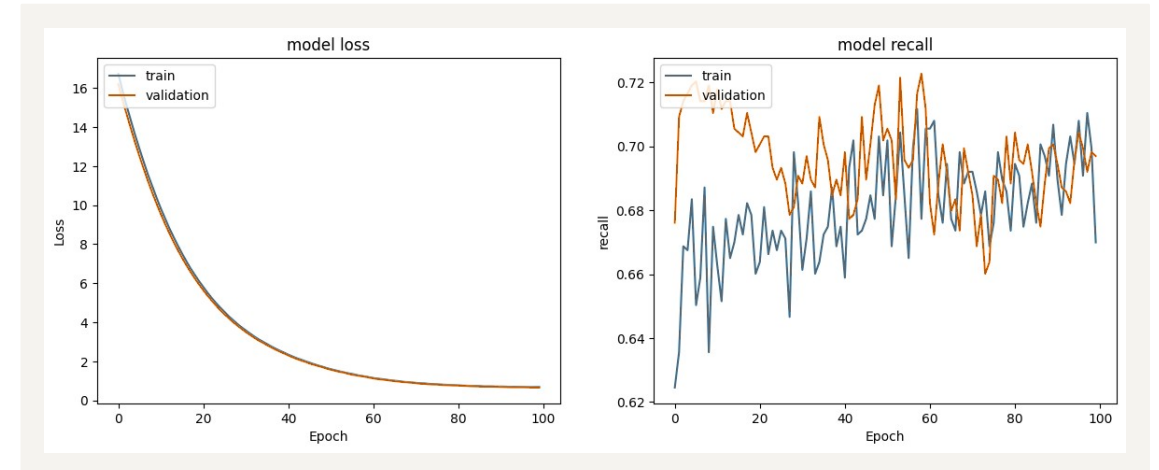
tanh vs ReLU

- With tanh, validation recall plateaus at 0.63 and training never catches up. ReLU reaches 0.74 with train and validation tracking together.



L2 vs L1

- L2 matches plain ReLU (0.74) with slightly smoother curves. L1 costs four points and adds noise.



Data background — 10,000 customers, 14 columns

Identifier columns (RowNumber, CustomerId, Surname) removed; Geography (France, Germany, Spain) and Gender encoded as dummies.

Variable	Range	Center	Reading
CreditScore	350 – 850	mean $\approx$ 650, sd 96.7	moderate variability within normal bounds
Age	18 – 92	mean $\approx$ 39	right-skewed; younger customer base
Tenure	0 – 10 years	mean $\approx$ 5, sd 2.9	even spread
Balance	0 – 250,898	mean $\approx$ 76,485, sd 62,397	many zero balances; wide spread
NumOfProducts	1 – 4	mean 1.5	most hold one or two
HasCrCard	0 / 1	70% hold a card	
IsActiveMember	0 / 1	51% active	slight majority
EstimatedSalary	11.58 – 199,992	mean $\approx$ 100,092, sd 57,510	minimum looks like an entry error
Exited	0 / 1	20.4% exited	target



Thank you

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